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United States Patent	12384250
Kind Code	B2
Date of Patent	August 12, 2025
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Torque-equalizing fault response for electric vehicle

Abstract

Various disclosed embodiments include illustrative controllers, dual power inverter modules, and electric vehicles. In an illustrative embodiment, a controller includes one or more processors associated with a first and second power inverter for the drive unit. Computer-readable media for the one or more processors are each configured to store computer-executable instructions configured to cause the one or more processors to apply a same fault action to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

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Appl. No.: 18/208999

Filed: June 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230322085 A1	Oct. 12, 2023

Related U.S. Application Data

continuation parent-doc US 17389058 20210729 US 11712967 child-doc US 18208999

Publication Classification

Int. Cl.: B60L3/00 (20190101); B60L50/51 (20190101)

U.S. Cl.:

CPC B60L3/003 (20130101); B60L3/0092 (20130101); B60L50/51 (20190201); B60L2210/40 (20130101); B60L2240/421 (20130101)

Field of Classification Search

CPC: B60L (3/00); B60L (50/51); B60L (3/0092); B60L (2240/421); B60L (2210/40)

USPC: 318/801

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present disclosure is a continuation of co-pending U.S. patent application Ser. No. 17/389,058, filed on Jul. 29, 2021, and entitled “TORQUE-EQUALIZING FAULT RESPONSE FOR ELECTRIC VEHICLE,” the contents of which are incorporated in full by reference herein.

INTRODUCTION

(1) The present disclosure relates to inverters for drive units of electric vehicles. A conventional electric vehicle with two electric motors per drive unit has two inverters (one inverter per motor) and each inverter is controlled by its own inverter controller. Use of a separate inverter controller for each inverter adds weight and increases energy consumption, thereby resulting in reduced vehicle range.

BRIEF SUMMARY

(2) Various disclosed embodiments include illustrative controllers, dual power inverter modules, and electric vehicles.

(3) In an illustrative embodiment, a controller includes one or more processors associated with a first and second power inverter for a drive unit for an electric vehicle. One or more computer-readable media for the one or more processors are each configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action to the first power inverter and the second power responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

(4) In another illustrative embodiment, a dual power inverter module includes a DC link capacitor electrically connectable to a source of high voltage direct current (DC) electrical power. A first power inverter is electrically connectable to the DC link capacitor and is configured to convert high voltage DC electrical power to three phase high voltage alternating current (AC) electrical power. The first power inverter is further configured to supply the three phase high voltage AC electrical power to a first electric motor. A second power inverter is electrically connectable to the DC link capacitor and is configured to convert high voltage DC electrical power to three phase high voltage AC electrical power. The second power inverter is further configured to supply the three phase high voltage AC electrical power to a second electric motor. A controller includes one or more processors associated with a first and second power inverter for a drive unit for an electric vehicle. One or more computer-readable media for the first processor and second computer-readable media are each configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

(5) In another illustrative embodiment, an electric vehicle includes a vehicle body, a high voltage direct current (DC) electrical battery disposed within the vehicle body, first and second (e.g., left and right) electric motors mechanically couplable to rotate at least one set of axles (e.g., coupled to wheels), and at least one dual power inverter module. The at least one dual power inverter module includes: a DC link capacitor electrically connectable to the high voltage DC electrical battery; a first power inverter electrically connectable to the DC link capacitor and configured to convert high voltage DC electrical power to three phase high voltage alternating current (AC) electrical power, the first power inverter being further configured to supply the three phase high voltage AC electrical power to an electric motor chosen from first and second (e.g., left and right) electric motors; a second power inverter electrically connectable to the DC link capacitor and configured to convert high voltage DC electrical power to three phase high voltage AC electrical power, the second power inverter being further configured to supply the three phase high voltage AC electrical power to the other electric motor chosen from the left and right electric motors; and a common controller electrically connectable to the first power inverter and the second power inverter. The common controller is configured to control the first power inverter and the second power inverter. The common controller includes one or more processors associated with the first power inverter and the second power inverter. One or more computer-readable media for the first processor and the second processor are each configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

(6) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Illustrative embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.

(2) FIG. 1A is a schematic illustration of an illustrative electric vehicle with at least one drive unit.

(3) FIG. 1B is a perspective view of lower body structure of the electric vehicle of FIG. 1 with illustrative drive units.

(4) FIG. 2A is a perspective view of an illustrative drive unit of FIG. 1B.

(5) FIG. 2B is another perspective view of the drive unit of FIG. 1B.

(6) FIG. 2C is a side plan view of the drive unit of FIG. 1B.

(7) FIG. 2D is an exploded perspective view of the drive unit of FIG. 1B.

(8) FIG. 2E is a perspective view of another illustrative drive unit.

(9) FIG. 3A is a block diagram in partial schematic form of an illustrative dual inverter with common control.

(10) FIG. 3B is a perspective view of an illustrative dual inverter with common control.

(11) FIG. 4A is a simplified schematic diagram of an illustrative dual inverter with a common DC link capacitor.

(12) FIGS. 4B and 4C are graphs of ripple waveforms with no cancellation.

(13) FIG. 4D is a flow chart of an illustrative method for synchronizing pulse width modulation clocks.

- (14) FIGS. 4E and 4F are graphs of illustrative ripple waveforms with cancellation.
- (15) FIG. 4G illustrates a look-up table.
- (16) FIG. 5A is a block diagram of illustrative components configured to place both inverters of a drive unit in a safe state responsive to a detected fault associated with either inverter.
- (17) FIG. 5B is a graph of torque versus speed.
- (18) FIG. 5C is a block diagram of details of components of FIG. 5A.
- (19) FIG. 5D is a simplified schematic diagram of open upper and lower banks of three-terminal power semiconductor devices.
- (20) FIG. 5E is a simplified schematic diagram of a shorted lower bank of three-terminal power semiconductor devices.
- (21) FIG. 5F is a simplified schematic diagram of a shorted upper bank of three-terminal power semiconductor devices.
- (22) FIG. 5G is a flow chart of an illustrative method of placing both inverters of a drive unit in a safe state responsive to a detected fault associated with either inverter.
- (23) FIG. 6A is a schematic diagram of an illustrative circuit for detecting loss of low voltage DC electrical power.
- (24) FIG. 6B is a block diagram in partial schematic form of details of an illustrative backup circuit for providing low voltage DC electrical power.
- (25) FIG. 6C is a flowchart of a method of placing both inverters of a drive unit in a safe state responsive to loss of low voltage DC electrical power.
- (26) Like reference symbols in the various drawings generally indicate like elements.

DETAILED DESCRIPTION

- (27) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.
- (28) Various disclosed embodiments include illustrative dual power inverter modules, electric vehicles, and methods.
- (29) Referring now to FIGS. 1A and 1B and given by way of overview, in various embodiments an electric vehicle **10** includes a vehicle body **12**. A high voltage direct current (DC) electrical battery **14** is disposed within the vehicle body **12**. Left and right front wheels **16** (only the left front wheel **16** is shown) and left and right rear wheels **18** (only the left rear wheel **18** is shown) are configured to rotate. At least one drive unit **20** is mechanically couplable to rotate the front wheels **16** or the rear wheels **18** (and in some embodiments one drive unit **20** may be mechanically couplable to rotate the front wheels **16** and another drive unit **20** may be mechanically couplable to rotate the rear wheels **18**). Each drive unit **20** is electrically connectable to receive high voltage DC electrical power from the battery **14**. Each drive unit **20** includes axles **22A** and **22B** that are mechanically couplable to rotate an associated wheel **16** or **18** and electrical motors **24A** and **24B** mechanically couplable to rotate its associated axle **22A** and **22B**, respectively. As will be explained below, each drive unit **20** also includes a dual power inverter module **26** that is electrically connectable to receive the high voltage DC electrical power from the battery **14**. The dual power inverter module **26** includes two inverters (not shown) configured to create three-phase, high voltage alternating current (AC) electrical power from the high voltage DC electrical power and provide the three-phase, high voltage AC electrical power to an associated electrical motor. A common controller (not shown) is configured to control both of the inverters in the dual power inverter module **26**.
- (30) For sake of brevity, illustrative details are set forth below by way of non-limiting examples in the context of a motor vehicle. However, it will be appreciated that the vehicle **10** can be any type of vehicle whatsoever as desired without limitation. Given by way of non-limiting example, in

various embodiments the vehicle **10** may be an electric vehicle (that is, an all-electrically driven vehicle) or a hybrid vehicle. For example and given by way of non-limiting examples, in various embodiments the vehicle **10** may include a motor vehicle driven by wheels and/or tracks, such as, without limitation, an automobile, a truck, a sport utility vehicle (SUV), a van, an all-terrain vehicle (ATV), a motorcycle, an electric bicycle, a tractor, a lawn mower such as without limitation a riding lawn mower, a snowmobile, and the like. Given by way of further non-limiting examples, in various embodiments the vehicle **10** may include a marine vessel such as, without limitation, a boat, a ship, a submarine, a submersible, an autonomous underwater vehicle (AUV), and the like. Given by way of further non-limiting examples, in various embodiments the vehicle **10** may include an aircraft such as, without limitation, a fixed wing aircraft, a rotary wing aircraft, and a lighter-than-air (LTA) craft.

(31) Also for sake of brevity, illustrative details regarding the drive unit **20** are set forth in the context of a motor vehicle. Because the vehicle **10** is not limited to the illustrative example of a motor vehicle, it will be appreciated that the drive unit **20** also is not limited to applicability to a motor vehicle. To that end, in various embodiments the motor (or motors) of the drive unit **20** are configured to drive the vehicle **10**. That is, in various embodiments the electric motor (or motors) of the drive unit **20** may drive any drive member that drives any propulsion device, such as without limitation a wheel or wheels, a track or tracks, a propeller or propellers, a propulsor or propulsors, a rotor or rotors, or the like, associated with the vehicle **10**.

(32) For example, in some embodiments in a motor vehicle one drive unit **20** may include one motor configured to drive one drive member such as an axle or a chain ring that drives one wheel or track, in some other embodiments in a motor vehicle one drive unit **20** may include one motor configured to drive an axle that rotates two wheels or two tracks, and in some other embodiments in a motor vehicle one drive unit **20** may include one motor configured to drive an axle that rotates one wheel or one track and another motor configured to drive another axle that rotates another wheel or another track.

(33) Similarly, in some embodiments in a marine vessel one drive unit **20** may include one motor configured to drive one propeller or propulsor, in some other embodiments in a marine vessel one drive unit **20** may include one motor configured to drive a shaft that rotates two propellers or two propulsors, and in some other embodiments in a marine vessel one drive unit **20** may include one motor configured to drive a shaft that rotates one propeller or propulsor and another motor configured to drive another shaft that rotates another propeller or propulsor.

(34) Likewise, in some embodiments in an aircraft one drive unit **20** may include one motor configured to drive one propeller or rotor, in some other embodiments in an aircraft one drive unit **20** may include one motor configured to drive a shaft that rotates two propellers or two rotors, and in some other embodiments in an aircraft one drive unit **20** may include one motor configured to drive a shaft that rotates one propeller or rotor and another motor configured to drive another shaft that rotates another propeller or rotor.

(35) Now that an overview has been set forth, illustrative details will be explained with examples that are given by way of illustration only and not of limitation.

(36) As mentioned above, the at least one drive unit **20** is mechanically couplable to rotate the front wheels **16** or the rear wheels **18** (and in some embodiments one drive unit **20** may be mechanically couplable to rotate the front wheels **16** and another drive unit **20** may be mechanically couplable to rotate the rear wheels **18**). As also mentioned above, each drive unit **20** includes the axles **22A** and **22B** that are mechanically couplable to rotate an associated wheel **16** or **18** and electrical motors **24A** and **24B** that are mechanically couplable to rotate its associated axle **22A** and **22B**, respectively.

(37) Referring additionally to FIGS. **2A-2E**, in various embodiments each electrical motor **24A** and **24B** is mechanically couplable to rotate its associated axle **22A** and **22B**, respectively, via a set of gears **28A** and **28B**, respectively. Each set of gears **28A** and **28B** is configured to provide speed and

torque conversions from its associated electrical motor **24A** or **24B**, respectively, to its associated axle **22A** or **22B**, respectively, and, ultimately, an associated wheel **16** or **18**.

(38) As shown in FIG. 2D, a shaft **21A** is configured to be rotated by its associated rotor **27** of its electrical motor **24A** and a shaft **21B** is configured to be rotated by its associated rotor **27** of its electrical motor **24B**. The shafts **21A** and **21B** are supported and constrained within bearings (not shown) disposed on a frame **19**. The set of gears **28A** is disposed within the frame **19** and is configured to rotatably engage the axle **22A** and the set of gears **28B** is disposed within the frame **19** and is configured to rotatably engage the axle **22B**. The electrical motor **24A** is configured to rotatably engage the set of gears **28A** and the electrical motor **24B** is configured to rotatably engage the set of gears **28B**.

(39) In various embodiments, each of the sets of gears **28A** and **28B** is configured to provide speed and torque conversions from its associated electrical motor **24A** or **24B** to its associated axle **22A** or **22B** and, ultimately, an associated wheel **16** or **18**. For example and given by way of illustration only and not of limitation, in various embodiments a gear **28C** is configured to be rotated by its associated shaft **21A** or **21B**. A gear **28D** is mounted on a shaft (not shown for purposes of clarity) and is configured to be meshedly engaged by the gear **28C**. A gear **28E** also is mounted on the shaft (not shown for purposes of clarity). A gear **28F** is mounted on the axle **22A** or **22B** and is configured to be meshedly engaged by the gear **28E**. It will be appreciated that, in various embodiments, the sets of gears **28A** and **28B** may include any number of suitable gears (such as, without limitation, planetary gears) with gear ratios selected as desired for a particular application to achieve desired speed and torque conversions. It will be appreciated that gears for use in electric vehicles are well known in the art. Therefore, further description of their construction and operation are not necessary for an understanding of disclosed subject matter.

(40) In some embodiments, the axles **22A** and **22B** may be fixedly couplable to their associated wheels **16** or **18**. For example and without limitation, in some such embodiments the front wheels **16** may be fixedly couplable to their associated axles **22A** and **22B**. It will be appreciated that such fixed coupling may help contribute to reducing mechanical complexity and may help contribute to enabling the front wheels **16** to remain steerable (such as when the vehicle **10** is being towed with the front wheels **16** and the rear wheels **18** engaged on a surface of a road (that is, flat towing)).

(41) In some other embodiments, the axles **22A** and **22B** may be removably couplable to their associated wheels **16** or **18**. For example and without limitation, in some such embodiments the rear wheels **18** may be removably couplable to their associated axles **22A** and **22B**. It will be appreciated that such removable coupling of the rear wheels **18** may help contribute to avoiding generation of braking torque and/or uncontrolled electrical generation during towing.

(42) It will be appreciated that each drive unit **20** drives either left and right front wheels **16** or left and right rear wheels **18**. Therefore, both of the electrical motors **24A** and **24B** of a given drive unit **20** may experience a same or similar range of speed and torque demands. To service such ranges of speed and torque demands, in various embodiments both of the electrical motors **24A** and **24B** of a given drive unit **20** may have the same voltage and current ratings. For example, in various embodiments (such as in high voltage systems), a voltage rating may be in a range from around 300 Vrms to around 600 Vrms line-to-line and a current rating may be in a range from around 300 Arms to around 900 Arms. Given by way of non-limiting example by way of illustration only, an illustrative voltage rating may be 312 Vrms and an illustrative current rating may be 550 Arms (based on a 400 VDC system). However, it will be appreciated that the electrical motors **24A** and **24B** may have any voltage ratings and any current ratings as desired for a particular application.

(43) In various embodiments, the electrical motors **24A** and **24B** may be any suitable type of electrical motor as desired. For example, in some embodiments the electrical motors **24A** and **24B** may include synchronous electrical motors. In some such embodiments, the synchronous electrical motors may include without limitation permanent magnet electrical motors or the like. In some other embodiments the electrical motors **24A** and **24B** may include without limitation an

asynchronous motor (or induction motor)—like a polyphase AC induction motor or the like.

(44) As mentioned above, in various embodiments each drive unit **20** drives either left and right front wheels **16** or left and right rear wheels **18** and, therefore, both of the electrical motors **24A** and **24B** of a given drive unit **20** may experience a same or similar range of speed and torque demands. Therefore, in various embodiments both inverters of a given drive unit **20** may have the same voltage output ratings and the same current output ratings. Given by way of illustration only and not of limitation, in various embodiments both inverters of a given drive unit **20** may have a voltage output rating of 312 Vrms and a current output rating of 550 Arms (based on 400 VDC input). However, it will be appreciated that the inverters of a given drive unit **20** may have any voltage output ratings and any current output ratings as desired for a particular application.

(45) It will be appreciated that, in various embodiments, the dual power inverter module **26** may be physically associated with the frame **19** in any suitable manner as desired for a particular application. For example and without limitation, in some embodiments and as shown in FIGS. 2A-2C the dual power inverter module **26** may be a module within a sealed container and physically disposed on the frame **19** external to the frame. As another example and as shown in FIG. 2E without limitation, in some other embodiments the dual power inverter module **26** may be integratably mountable with the frame **19**. In some such embodiments a housing **29** has an open face (not shown) defined therein. Inverter circuitry (discussed below) is disposed in the housing **29**. In such embodiments, the open face of the housing **29** is mated to an opening (not shown) in the frame **19**. Such other embodiments are discussed in commonly-owned U.S. patent application Ser. No. 17/244,288 filed Apr. 29, 2021 entitled “INVERTER MODULE INTEGRATABLY MOUNTABLE WITH DRIVE UNIT OF VEHICLE” assigned to and filed by Applicant, the entire contents of which are hereby incorporated by reference.

(46) Referring additionally to FIGS. 3A and 3B, in various embodiments a dual power inverter module (DPIM) **26** is provided. As mentioned above, in various embodiments the DPIM **26** includes two inverters **30A** and **30B** configured to create three-phase, high voltage AC electrical power from the high voltage DC electrical power and provide the three-phase, high voltage AC electrical power to an associated electrical motor **24A** or **24B** and a common controller **32** configured to control both of the inverters **30A** and **30B**.

(47) In various embodiments, a DC link capacitor **34** is electrically connectable to a source of high voltage DC electrical power, such as the battery **14**. In some embodiments, an electrical connection to the battery **14** may include an electrical connection **36**. Suitable electrical cables **38** may be electrically connected to the electrical connection **36** and may be electrically connectable to the electrical battery **14**.

(48) In various embodiments, a power inverter **30A** is electrically connectable to the DC link capacitor **34** and is configured to convert high voltage DC electrical power to three phase high voltage AC electrical power. The power inverter **30A** is further configured to supply the three phase high voltage AC electrical power to an electric motor **24A**, such as a right motor or a left motor. A power inverter **30B** is electrically connectable to the DC link capacitor **34** and configured to convert high voltage DC electrical power to three phase high voltage AC electrical power. The power inverter **30B** is further configured to supply the three phase high voltage AC electrical power to an electric motor **24B**, such as the other of the left motor or right motor.

(49) In various embodiments the common controller **32** is electrically connectable to the power inverter **30A** and the power inverter **30B**. The common controller **32** is configured to control the power inverter **30A** and the power inverter **30B**. The common controller **32** may be any suitable computer processor-based controller as desired. Given by way of example only and not of limitation, in various embodiments the common controller **32** may include a computer processing unit (CPU), a general purpose processor, a digital signal processor, a field programmable gate array, or the like, and/or any combination thereof. While controllers are well known and further description of their construction and operation are not necessary for an understanding of disclosed

subject matter, further details regarding the common controller **32** will be set forth below regarding additional functions.

(50) In various embodiments, the common controller **32** is electrically connectable to receive low voltage DC electrical power, such as 12 VDC. The common controller **32** also is electrically connectable to receive vehicle status signals and vehicle fault indication signals. Illustrative responses to various vehicle faults and to loss of 12 VDC will be discussed further below.

(51) In various embodiments, the power inverter **30A** and the power inverter **30B** each include a bank **40** of three-terminal power semiconductor devices **42** and a bank **44** of the three-terminal power semiconductor devices **42**.

(52) In some embodiments, the three-terminal power semiconductor devices **42** may include insulated gate bipolar transistors (IGBTs). In some such embodiments, the IGBTs may include silicon (Si) IGBTs. In some embodiments, the three-terminal power semiconductor devices **42** may include metal-oxide-semiconductor field effect transistors (MOSFETs). In some such embodiments, the MOSFETs may include silicon carbide (SiC) MOSFETs. However, it will be appreciated that the three-terminal power semiconductor devices **42** may also include power semiconductor devices **42** that include at least three terminals and may include additional terminals—such as, for example and without limitation, Kelvin source terminal, Kelvin emitter terminal, current sense terminal, and/or temperature sense terminal.

(53) It will be appreciated that SiC MOSFETs may offer advantages over Si IGBTs at low phase currents and that SiC MOSFETs have lower conduction drop compared to IGBTs below 700 Apk. However, it will be appreciated that this value may change depending upon size of the inverter. As such, it will be appreciated that SiC MOSFETs may offer up to around 3-5% efficiency gains over a typical drive cycle, as compared to Si IGBTs.

(54) In some all-wheel drive (AWD) electric vehicles **10**, the front wheels **16** function as the only drive wheels until additional torque and/or power beyond a predetermined amount is entailed. In such electric vehicles **10**, the rear wheels are only driven by their associated drive unit **20** when torque and/or power beyond the predetermined amount is to be delivered. In some such AWD electric vehicles **10**, to take advantage of efficiencies of SiC MOSFETs over Si IGBTs the drive unit **20** that drives the front wheels **16** may include inverters **40** that include SiC MOSFETs and the drive unit **20** that drives the rear wheels **18** may include inverters **40** that include Si IGBTs. However, it will be appreciated that any of the drive units **20** may include inverters **40** that include either SiC MOSFETs or Si IGBTs as desired.

(55) In various embodiments the power inverter **30A** includes a gate drive circuit **46A** configured to drive gate terminals **48** of the banks **40** and **44** of three-terminal power semiconductor devices **42** of the power inverter **30A**. Similarly, the power inverter **30B** includes a gate drive circuit **46B** configured to drive gate terminals **48** of the banks **40** and **44** of three-terminal power semiconductor devices **42** of the power inverter **30B**. In various embodiments the controller **32** is configured to, among other functions, generate low-power turn on and turn off signals **66A** and **66B** and provide the turn on and turn off signals **66A** and **66B** to the gate drive circuits **46A** and **46B**, respectively. The low-power turn on and turn off signals **66A** and **66B** may be on the order of a few milliamperes of electrical current and logic level voltages such as 3.3 or 5V.

(56) In various embodiments the gate drive circuits **46A** and **46B** include suitable power amplifiers that amplify the low-power turn on and turn off signals **66A** and **66B** and generate high-power turn on and turn off signals **66A'** and **66B'**. To drive the gate terminals **48**, the high-power turn on and turn off signals **66A'** and **66B'** may be on the order of several hundreds of milliamperes of electrical current or on the order of amperes of electrical current, with voltages in the range of 15-20V, as desired for a particular application. The high-power turn on and turn off signals **66A'** and **66B'** are, in turn, electrically coupled to drive associated gate terminals **48**. Gate drive circuits are well known and further description of their construction and operation are not necessary for an understanding of disclosed subject matter.

(57) In view of the illustrative details provided above by way of non-limiting examples, it will be appreciated that, in various embodiments, provision of one controller **32** for two inverters **30A** and **30B** can provide for use of: (i) a single interface for vehicle status signals and vehicle fault indication signals; (ii) a single DC link capacitor **34**; (iii) a single common controller **32**; and (iv) a single interface for low voltage DC electrical power (such as 12 VDC).

(58) In view of the illustrative details provided above by way of non-limiting examples, it will be appreciated that, in various embodiments, integrating the mounting of the DPIM **26** with the drive unit **20** can provide for integration of coolant interfaces. For example, in various embodiments water cooling provided for the inverters **30A** and **30B** can be provided to stator windings of the electrical motors **24A** and **24B**.

(59) As discussed above, in various embodiments a single motor **24A** or **24B** drives a single wheel, such as one front wheel **16** or one rear wheel **18**. It will be appreciated that each wheel can operate at a unique speed and torque. Given by way of non-limiting examples, going around a curve or loss of traction can create different wheel speeds and traction control or torque vectoring can result in different wheel torques. As also discussed above, in various embodiments the dual power inverter module **26** combines two inverters **30A** and **30B** into one module **26** and, as such, shares common components, such as the DC link capacitor **34**.

(60) To that end, various embodiments include only one DC link capacitor **34** that is electrically connectable to provide high voltage DC electrical power to both inverters **30A** and **30B** of the DPIM **26**. It will be appreciated that it may be desirable to reduce and possibly minimize size of the DC link capacitor **34**. As explained below, various embodiments can help contribute to reducing stress on the DC link capacitor **34** (that may have a reduced size) and ripple current on the DC high voltage bus (that may include the electrical cables **38**) due to generation of high-frequency current harmonics from the inverters **30A** and **30B**.

(61) In various embodiments, the inverters **30A** and **30B** use pulse width modulation (PWM) to create a variable amplitude and frequency voltage source to drive the electric motors **24A** and **24B**. Different PWM methods (such as, for example, continuous PWM and discontinuous PWM) can be employed as desired for a particular situation. Each of the PWM methods creates its own unique harmonic spectrum of ripple current that may be reflected on the DC bus.

(62) For example, in continuous PWM each phase is switching continuously (that is, in various embodiments all of the three-terminal power semiconductor devices **42** in both of the inverters **30A** and **30B** are switching continuously). As a result, continuous PWM can result in a not-insignificant amount of switching losses in the inverters **30A** and **30B**. In continuous PWM, the second harmonic of the switching frequency is the dominant harmonic frequency in the ripple current. Continuous PWM (such as space vector modulation) can be used in situations such as when it is desired to minimize harmonic content and ripple on AC output and DC input currents, as well as minimize acoustic noise. For example, some vehicles might use continuous PWM at high torque to minimize acoustic noise.

(63) As another example, in discontinuous PWM each phase is not switching continuously (that is, in various embodiments all of the three-terminal power semiconductor devices **42** in both of the inverters **30A** and **30B** are not switching continuously). In such embodiments, only two of the three phases are switching at any one time, and the remaining third phase has either the upper or lower switch turned on continuously, thereby helping to contribute to increasing inverter efficiency and helping to contribute to reducing losses during discontinuous PWM. It will be appreciated that use of discontinuous PWM may have other non-beneficial effects, such as increased acoustic noise or harmonic content in the AC output or DC input currents. In discontinuous PWM each phase has two 60 degree segments where the switches are held either low or high, and is not switching for a total of 120 degrees over a fundamental period. That is, in discontinuous PWM each phase is not switching one-third of the time. This results in significantly lower switching losses and higher efficiency. It will be appreciated that, while discontinuous PWM entails lower switching losses

than those entailed in continuous PWM, discontinuous PWM may entail higher acoustic noise than that associated with continuous PWM and harmonics may be placed onto the motors **24A** and **24B** as well as the DC link capacitor **34**. In discontinuous PWM, the first harmonic of the switching frequency is the dominant harmonic frequency. Discontinuous PWM can be used in situations when the highest efficiency and lowest losses are desired, and acoustic noise is not problematic. As an example, some vehicles might have thermal problems at high torque and, as a result, use discontinuous PWM to reduce losses under those conditions.

(64) In various embodiments and referring additionally to FIG. **4A**, the DC link capacitor **34** is used to decouple effects of inductance L_{cable} from the DC voltage source (that is, the electrical battery **14**) to the inverters **30A** and **30B**. Referring additionally to FIGS. **4B** and **4C**, the DC link capacitor **34** provides a low impedance path for ripple currents **50** that are generated by the inverters **30A** and **30B** and might otherwise flow back onto the DC high voltage bus. The ripple currents **50** are a factor in sizing the DC link capacitor **34** and, as a result, reducing the ripple currents **50** can help contribute to reducing size of the DC link capacitor **34** (and stress on the DC link capacitor **34**). The ripple currents **50** are a result of the AC load current flowing into the motors **24A** and **24B** and pulse width modulation (PWM) of the inverters **30A** and **30B**. As shown in FIGS. **4B** and **4C** and given by way of illustration only and not of limitation, with a motor current of 550 Arms, a modulation index of 0.48, a power factor of 1, a switching frequency of 10 KHz, and a fundamental frequency of 250 Hz, without cancellation the ripple currents **50** can have an amplitude of around 712 Arms.

(65) Referring additionally to FIG. **4D**, in various embodiments a method **52** is provided for synchronizing pulse width modulation clocks. It will be appreciated that, in various embodiments, synchronizing pulse width modulation clocks can help contribute to cancelling the ripple currents **50**. The method **52** begins at a block **53**. At a block **54** a pulse width modulation method of a first power inverter and a pulse width modulation method of a second power inverter are identified. At a block **56**, a switching frequency of the first power inverter and a switching frequency of the second power inverter are identified and compared. At a block **58**, an optimized phase shift between the first power inverter and the second power inverter is determined responsive to the pulse width modulation method of the first power inverter and the pulse width modulation method of the second power inverter and the switching frequency of the first power inverter and the switching frequency of the second power inverter. At a block **60** the optimized phase shift is synchronized between the first power inverter and the second power inverter.

(66) The method **52** ends at a block **61**.

(67) As shown in FIGS. **4E** and **4F** and given by way of illustration only and not of limitation, with a motor current of 550 Arms, a modulation index of 0.48, a power factor of 1, a switching frequency of 10 KHz, and a fundamental frequency of 250 Hz, with cancellation as described herein the ripple currents **50** can be reduced to have an amplitude of around 147 Arms.

(68) With the above overview in mind, in various embodiments and as shown in FIG. **3A** the common controller **32** includes a processor **64A** and a processor **64B**. In some embodiments the processors **64A** and **64B** may be separate processors. However, it will be appreciated that in some other embodiments (that use two PWM generators) the functions of the processors **64A** and **64B** may be combined into a single processor (e.g., using multitasking, multithreading, time slicing, and the like) as desired for a particular application. Regardless of the processors **64A** and **64B** being separate or combined into a single processor, it will be appreciated that the processor **64A** functions as a Master (or in the case of a single processor, the functions of processor **64A** function as Master processes) and the processor **64B** functions as a Slave (or in the case of a single processor, the functions of processor **64B** function as Slave processes).

(69) The processor **64A** is operably coupled to computer-readable media **65A**, such as any suitable computer memory, configured to store computer-executable instructions configured to cause the processor **64A** to perform functions described below. The processor **64B** is operably coupled to

computer-readable media 65B, such as any suitable computer memory, configured to store computer-executable instructions configured to cause the processor 64B to perform functions described below. The processor 64A is configured to generate a first clock signal (e.g., using a crystal) for controlling generation of the turn on and turn off signal 66A for driving the power inverter 30A and a second clock signal. The second clock signal is provided to the processor 64B for controlling generation of the turn on and turn off signal 66B for driving the power inverter 30B. (70) The processor 64A selects a PWM method for the power inverter 30A and the processor 64B selects a PWM method for the power inverter 30B. The processor 64B (that is, the Slave) informs the processor 64A (that is, the Master) of its PWM method (except for cases where switching of PWM methods is not employed). Factors for selection of continuous PWM and discontinuous PWM have been discussed above.

(71) The processor 64A selects a switching frequency for the power inverter 30A and the processor 64B selects a switching frequency for the power inverter 30B. The frequencies are even multiples of each other, and are selected from predetermined values. Given by way of illustration only and not of limitation, in various embodiments the switching frequencies may be 2.5 KHz and 10 KHz, thereby resulting in an even multiple of four. However, it will be appreciated that other frequencies may be selected as desired (that result in even multiples). In various embodiments the processor 64A (that is, the Master) puts out a reference that is at the lowest selectable frequency, with the desired phase shift as a function of the PWM modes.

(72) The processor 64A is configured to determine an optimized phase shift between the power inverter 30A and the power inverter 30B responsive to the PWM method of the power inverter 30A and the PWM method of the power inverter 30B and the switching frequency of the power inverter 30A and the switching frequency of the power inverter 30B. For example, in some such embodiments and referring additionally to FIG. 4G, the processor 64A may access a look-up-table 68 that is populated with cells that include values of optimized phase shift arranged according to rows 70 of PWM method and columns 72 of PWM method. It will be appreciated that, in some embodiments, the PWM method of the power inverter 30A and the PWM method of the power inverter 30B may be the same pulse width modulation method. It will also be appreciated that, in some other embodiments, the PWM method of the power inverter 30A and the PWM method of the power inverter 30B may be different PWM methods.

(73) As shown in FIG. 4G, when both of the power inverters 30A and 30B use continuous PWM, the phase shift is set to 90 degrees. When both of the power inverters 30A and 30B use discontinuous PWM, the phase shift is set to 180 degrees. In the event of different PWM methods, the phase is set to 90 degrees.

(74) In some other such embodiments, the processor 64A may execute an algorithm for determining an optimized phase shift between the power inverter 30A and the power inverter 30B responsive to the PWM method of the power inverter 30A and the PWM method of the power inverter 30B. For example, the algorithm may include if-then statements, such as: if both PWM methods are continuous PWM, then the optimized phase shift is 90 degrees; if both PWM methods are discontinuous PWM, then the optimized phase shift is 180 degrees; and if one PWM method is continuous PWM and another PWM method is discontinuous PWM, then the optimized phase shift is 90 degrees.

(75) The processor 64A is also configured to synchronize the optimized phase shift between the power inverter 30A and the power inverter 30B. As mentioned above, the processor 64B is configured to receive the second clock signal and to drive the power inverter 30B.

(76) In various embodiments, the processor 64A is further configured to shift the second clock signal from the first clock signal by the determined optimized phase shift.

(77) In various embodiments, the processor 64A is further configured to identify a dominant harmonic frequency among harmonic frequencies of the PWM method of the power inverter 30A and the PWM method of the power inverter 30B and to determine the optimized phase shift

between the power inverter **30A** and the power inverter **30B** responsive to the dominant harmonic frequency among the harmonic frequencies of the PWM method of the power inverter **30A** and the PWM method of the power inverter **30B**. It will be appreciated that determining the optimized phase shift between the power inverter **30A** and the power inverter **30B** responsive to the dominant harmonic frequency among the harmonic frequencies of the PWM method of the power inverter **30A** and the PWM method of the power inverter **30B** can help contribute to cancelling dominant high frequency components of the ripple currents **50**.

(78) In some such embodiments, the dominant harmonic frequency may include a second harmonic frequency (harmonic of the switching frequency). For example, in such embodiments the PWM method may include continuous PWM. In such embodiments, the optimized phase shift is 90 degrees. It will be appreciated that a phase shift of 90 degrees shifts the dominant harmonic (second harmonic frequency) by 180 degrees, thereby resulting in cancellation of the dominant harmonic component of the ripple current **50**.

(79) In some other such embodiments, the dominant harmonic frequency may include a first harmonic frequency. For example, in such embodiments the PWM method may include discontinuous PWM. In such embodiments, the optimized phase shift is 180 degrees. It will be appreciated that a phase shift of 180 degrees shifts the dominant harmonic (first harmonic frequency) by 180 degrees, thereby resulting in cancellation of the dominant harmonic component of the ripple current **50**.

(80) It will be appreciated that actual DC bus harmonics may reside in side band groups surrounding the switching frequency. The separation of these harmonics from the switching frequency harmonic is a function of the motor fundamental frequency.

(81) In various embodiments the processor **64A** is configured to identify and compare a switching frequency of the power inverter **30A** and a switching frequency of the power inverter **30B**. As discussed above, only two frequencies that are even multiples of each other are used.

(82) For example, in some embodiments it may be desirable to change switching frequency. For example, in some embodiments switching frequency may be 10 KHz. In such embodiments, at low motor speed (for example, below 500 RPM) it may be desirable to reduce switching frequency to 2.5 KHz to protect the switches **42** in the inverters **30A** and **30B**, thereby helping to reduce stresses in the inverters **30A** and **30B**. Maintaining even multiple relation between the switching frequencies allows for alignment of harmonics and increased opportunity for cancellation of dominant harmonics.

(83) Electric vehicles in which each wheel is independently driven by its own associated electrical motor via its own axle does not have any mechanical coupling between the wheels. In such vehicles, if one inverter shuts down due to a fault and the other inverter does not react appropriately, then a torque difference might exist between the two wheels. Such a resultant torque difference might have negative impact on controllability of the vehicle.

(84) To help avoid such a torque difference and referring additionally to FIGS. 5A-5G, in various embodiments a fault that is associated with either the power inverter **30A** or the power inverter **30B** (or circuitry described below that is associated with both the inverters **30A** and **30B**) results in application of a same fault action (described below) to both the inverters **30A** and **30B**. In such embodiments, application of the same fault action to both the inverters **30A** and **30B** puts both the inverters **30A** and **30B** in a “safe state” and helps contribute to applying an equalized torque to both of the wheels associated with the drive unit **20** that includes the inverters **30A** and **30B**. By applying the same fault action to both the inverters **30A** and **30B** and equalizing torque to both of the wheels associated with the drive unit **20** that includes the inverters **30A** and **30B**, various embodiments are able to help contribute to reducing the possibility of generation of torque difference that might have negative impact on controllability of the vehicle.

(85) As discussed above, the processor **64A** is operably coupled to computer-readable media **65A**, such as any suitable computer memory, configured to store computer-executable instructions

configured to cause the processor **64A** to perform functions described below. As also discussed above, the processor **64B** is operably coupled to computer-readable media **65B**, such as any suitable computer memory, configured to store computer-executable instructions configured to cause the processor **64B** to perform functions described below. In various embodiments and as will be described below, the computer-executable instructions are configured to cause its associated processor **64A** and **64B** to apply a same fault action to the power inverter **30A** and the power inverter **30B**, respectively, for applying equalized torque to each wheel **16** or **18** operatively coupled to the drive unit **20** responsive to a fault associated with the power inverter **30A** or the power inverter **30B**.

(86) In various embodiments and as shown in FIG. 5A, various faults associated with the inverters **30A** and **30B** are monitored by the processors **64A** and **64B** for the inverters **30A** and **30B**, respectively. For such faults, in various embodiments the same fault response is applied by the processors **64A** and **64B** to the three-terminal power semiconductor devices **42** (FIG. 3A) of both of the inverters **30A** and **30B**. In various embodiments, such faults associated with the inverters **30A** and **30B** and monitored by the processors **64A** and **64B**, respectively, may include, without limitation, overcurrent, overvoltage, undervoltage, over temperature, overspeed, and the like. In various embodiments, signals indicative of such faults may be provided to the controller **32** via a data link **74** that may include any suitable data communication connection or network as desired, such as without limitation a wide area network (WAN), a local area network (LAN), a controller area network (CAN), a peer-to-peer network, a data bus, or the like and provided to the processors **64A** and **64B**. In various embodiments, signals indicative of vehicle status, such as motor speed and voltage of the battery **14**, are also provided to the controller **32** via the data link **74**.

(87) In various embodiments and as also shown in FIG. 5A, the controller **32** includes a communications link **82** between the processors **64A** and **64B**. It will be appreciated that the communications link **82** can enable the processor **64A** or the processor **64B** to communicate to the other processor **64B** or **64A**, respectively, that a fault monitored by the processor **64A** or the processor **64B** has been detected, that a fault action is to be taken, and what fault action is to be taken. The communications link **82** may include any suitable data link or data bus as desired.

(88) In various embodiments, the fault action for a fault monitored by the processors **64A** and **64B** for the inverters **30A** and **30B**, respectively, may include an action such as opening all of the three-terminal semiconductor devices **42** of the inverters **30A** and **30B** and/or shorting the three-terminal semiconductor devices **42** of either the bank **40** or the bank **44** of the inverters **30A** and **30B**. As will be explained below, in various embodiments the fault action applied to both of the inverters **30A** and **30B** for a fault monitored by the processors **64A** and **64B** depends on speed of the motors. As will also be explained below, the fault action applied to both of the inverters **30A** and **30B** for a fault that is not monitored by the processors **64A** and **64B** does not depend on speed of the motors.

(89) In various embodiments and as mentioned above, in various embodiments the fault action for a fault monitored by the processors **64A** and **64B** for the inverters **30A** and **30B**, respectively, may include an action such as opening all of the three-terminal semiconductor devices **42** of the inverters **30A** and **30B** and/or shorting the three-terminal semiconductor devices **42** of either the bank **40** or the bank **44** of the inverters **30A** and **30B** and may depend on speed of the motors. For such faults that are monitored by the processors **64A** and **64B**, it will be appreciated that the fault action applied to both of the inverters **30A** and **30B** simultaneously can help contribute to reducing braking torque and reducing regenerative electrical current to the electrical battery **14**. Because the processors **64A** and **64B** are operatively coupled to various data communication connections or networks to receive data regarding speed of the motors, the processors **64A** and **64B** suitably are configured to determine the same fault action to be applied based on speed of the motors. In various embodiments, the processor **64A** selects the appropriate speed-dependent fault action for its associated electrical motor **24A** and the processor **64B** selects the appropriate speed-dependent fault action for its associated electrical motor **24B**. Fault actions for other faults—that are not

monitored by the processors **64A** and **64B**— do not depend on speed of the motors and are discussed further below.

(90) As shown in FIG. 5B, in various embodiments the same fault action to be applied the inverters **30A** and **30B** may be based on speed of the motors. In such embodiments the same fault action may include opening all of the three-terminal semiconductor devices **42** of the inverters **30A** and **30B** below a threshold speed $v_{sub.th}$ of the motor and shorting the three-terminal semiconductor devices **42** of either the bank **40** or the bank **44** of the inverters **30A** and **30B** above the threshold speed $v_{sub.th}$.

(91) As shown in FIG. 5B, a graph **76** plots speed of the motor versus torque for various conditions of the three-terminal semiconductor devices **42** of the inverters **30A** and **30B**. A curve **78** shows torque that results from opening all of the three-terminal semiconductor devices **42** of the inverters **30A** and **30B**. Below the threshold speed $v_{sub.th}$ torque is substantially insignificant and above the threshold speed $v_{sub.th}$ braking torque becomes increasingly significant with increased speed of the motors. It will also be appreciated that, in such instances, back electromotive force (back EMF) increases with increasing speed of the motors which can, in some instances, result in the electrical motors **24A** and **24B** possibly operating as uncontrolled generators that can generate back EMF and apply unwanted regenerated electrical current to the DC link capacitor **34** and the electrical battery **14**.

(92) As also shown in FIG. 5B, a curve **80** shows torque that results from a three-phase short of the three-terminal semiconductor devices **42** of either the bank **40** or the bank **44** of three-terminal semiconductor devices **42**. As speed of the motors increases from zero, braking torque rapidly increases and reaches a maximum value of braking torque. As speed of the motors continues to increase, braking torque decreases and approaches an asymptotic minimized value before the speed of the motors reaches the threshold speed $v_{sub.th}$.

(93) Thus, in various embodiments in which the fault is monitored by the processors **64A** and **64B** for the inverters **30A** and **30B**, the fault action suitably includes a speed-dependent fault action that can help contribute to simultaneously minimize braking torque and unwanted regenerative electrical current to the DC link capacitor **34** and the electrical battery **14**. In such embodiments, the fault action suitably includes opening all of the three-terminal power semiconductor devices **42** of both of the inverters **30A** and **30B** when the speed of the motors is less than the threshold speed $v_{sub.th}$ and shorting the three-terminal power semiconductor devices **42** of one bank of the three-terminal power semiconductor devices **42** (that is, either the bank **40** or the bank **44**) of the inverters **30A** and **30B** when the speed of the motors is greater than the threshold speed $v_{sub.th}$.

(94) In various embodiments, fault actions based on the curve **78** and the graph **80** may be implemented using back EMF and voltage of the battery **14** (as opposed to taking fault action directly in response to reported motor speed). For example, in various embodiments motor back EMF is computed and is compared to voltage of the battery. In such embodiments, when back EMF is less than battery voltage (by a design safety margin selected as desired), then a speed-dependent fault action includes opening all of the three-terminal power semiconductor devices **42** of both of the inverters **30A** and **30B**. When back EMF exceeds a predetermined percentage of battery voltage, then a speed-dependent fault action includes shorting the three-terminal power semiconductor devices **42** of one bank of the three-terminal power semiconductor devices **42** (that is, either the bank **40** or the bank **44**) of the inverters **30A** and **30B**. If desired, use of an amount of hysteresis can prevent “chattering” back-and-forth between different fault actions.

(95) As shown in FIG. 5A, the processor **64A** is associated with the power inverter **30A** and the processor **64B** is associated with the power inverter **30B**. The memory **65A** (that is, computer-readable media) and the memory **65B** (again, computer-readable media) each are configured to store computer-executable instructions configured to cause its associated processor **64A** and **64B**, respectively, to apply the same fault action to the power inverter **30A** and the power inverter **30B** for applying equalized torque to each wheel **16** or **18** operatively coupled to the drive unit **20**

responsive to a fault associated with the power inverter **30A** or the power inverter **30B**.

(96) As also shown in FIG. 5A, in various embodiments signals that are indicative of parameters such as vehicle speed, faults such as those described above, and the like are supplied to the controller **32** via the data link **74** and are provided to the processors **64A** and **64B**. In such embodiments the computer-executable instructions are further configured to cause its associated processor **64A** or **64B** to monitor for the fault.

(97) Low-voltage DC electrical power (such as 12V) is provided to the controller **32** for powering components as desired. The controller **32**, in turn, provides 12V DC electrical power to the gate drive circuits **46A** and **46B**.

(98) As shown in FIG. 5A, the processor **64A** is operatively coupled to provide a control signal **86** to drives **46A1** for the gates **48** of the bank **40** (sometimes referred to as an “upper bank”) and a control signal **88** to drives **46A2** for the gates **48** of the bank **44** (sometimes referred to as a “lower bank”). Similarly, the processor **64B** is operatively coupled to provide a control signal **90** to drives **46B1** for the gates **48** of the bank **40** (sometimes referred to as an “upper bank”) and a control signal **92** to drives **46B2** for the gates **48** of the bank **44** (sometimes referred to as a “lower bank”). In various embodiments the drives **46A1**, **46A2**, **46B1**, and **46B2** are suitable power amplifiers that amplify the low-power control signals **86**, **88**, **90**, and **92** and generate high-power, fault action signals **86'**, **88'**, **90'**, and **92'** that are, in turn provided to their associated gate terminals **48**.

(99) As shown in FIG. 5C, in various embodiments additional circuitry is entailed in providing the control signals **86**, **88**, **90**, and **92** to their respective drives **46A1**, **46A2**, **46B1**, and **46B2**. Each processor **64A** and **64B** is operatively coupled to receive the fault indication signal as discussed above, and the processors **64A** and **64B** are operatively coupled with each other via the communications link **82**. The processor **64A** is operatively coupled to provide the control signals **86** and **88** to buffers **94** and **96**, respectively, and the processor **64B** is operatively coupled to provide the control signals **90** and **92** to buffers **98** and **100**, respectively. The buffers **94**, **96**, **98**, and **100** are any suitable buffer, such as without limitation an octal buffer or the like.

(100) The buffer **94** is operatively coupled to the drives **46A1**, the buffer **96** is operatively coupled to the drives **46A2**, the buffer **98** is operatively coupled to the drives **46B1**, and the buffer **100** is operatively coupled to the drives **46B2**. When a monitored fault indication signal for an associated inverter **30A** or **30B** is received by either the processor **64A** or the processor **64B**, the processor that receives the monitored fault indication signal communicates existence of the monitored fault via the communications link **82**. Fault actions are taken as follows.

(101) When the processor **64A** receives a fault indication signal, the processor **64A** communicates to the processor **64B** via the communications link **82** that a fault monitored by the processor **64A** has been detected, that a fault action is to be taken by the processor **64B**, and (as discussed below) what fault action is to be taken by the processor **64B**. Conversely, when the processor **64B** receives a fault indication signal, the processor **64B** communicates to the processor **64A** via the communications link **82** that a fault monitored by the processor **64B** has been detected, that a fault action is to be taken by the processor **64A**, and (as discussed below) what fault action is to be taken by the processor **64A**. It will also be appreciated that in various embodiments the processor that receives the fault indication signal also performs the fault action (that it asks the other processor to take as well).

(102) When speed of the motors is less than the threshold speed v_{in} (or back EMF is less than voltage of the battery **14** by at least a design safety margin) and a fault has been detected, the processors **64A** and **64B** generate the control signals **86**, **88**, **90**, and **92** that are configured to turn off all the gate terminals **48** of the inverters **30A** and **30B** and, as a result, cause all of the three-terminal power semiconductor devices **42** of the inverters **30A** and **30B** to open. The control signals **86**, **88**, **90**, and **92** are provided to the buffers **94**, **96**, **98**, and **100**, respectively, and are, in turn, provided to the drives **46A1**, **46A2**, **46B1**, and **46B2** which provide the fault action signals **86'**, **88'**, **90'**, and **92'** to all the gate terminals **48** of the inverters **30A** and **30B**. As shown in FIG. 5D, all of

the three-terminal power semiconductor devices **42** in the banks **40** and **44** of the inverters **30A** and **30B** are caused to open.

(103) When speed of the motors is greater than the threshold speed with (or back EMF exceeds a predetermined percentage of battery voltage) and a fault has been detected, the processors **64A** and **64B** generate the control signals **86**, **88**, **90**, and **92** that are configured to turn off the gate terminals **48** of the three-terminal power semiconductor devices **42** in one of the banks **40** or **44** of the inverters **30A** and **30B** and turn on the gate terminals **48** of the three-terminal power semiconductor devices **42** in the other of the banks **44** or **40** of the inverters **30A** and **30B**. As a result, the three-terminal power semiconductor devices **42** of one of the banks **40** or **44** of both of the inverters **30A** and **30B** are caused to open and the three-terminal power semiconductor devices **42** of the other of the banks **44** or **40** of both of the inverters **30A** and **30B** are caused to short.

(104) It will be appreciated that the three-terminal power semiconductor devices **42** of either of the banks **40** or **44** of both of the inverters **30A** and **30B** may be caused to open or short as desired. It will also be appreciated that three-terminal power semiconductor devices **42** that have faulted will not be caused to open or short (because they have faulted). In some embodiments and as shown in FIG. 5E, the three-terminal power semiconductor devices **42** of the bank **40** of both of the inverters **30A** and **30B** are caused to open and the three-terminal power semiconductor devices **42** of the bank **44** of both of the inverters **30A** and **30B** are caused to short. In such embodiments, the processors **64A** and **64B** generate the control signals **86** and **90** that are configured to turn off all the gate terminals **48** in the bank **40** of the inverters **30A** and **30B** and, as a result, cause all of the three-terminal power semiconductor devices **42** in the bank **40** of the inverters **30A** and **30B** to open. The processors **64A** and **64B** also generate the control signals **88** and **92** that are configured to turn on all the gate terminals **48** in the bank **44** of the inverters **30A** and **30B** and, as a result, cause all of the three-terminal power semiconductor devices **42** in the bank **44** of the inverters **30A** and **30B** to short. The control signals **86**, **88**, **90**, and **92** are provided to the buffers **94**, **96**, **98**, and **100**, respectively, and are, in turn, provided to the drives **46A1**, **46A2**, **46B1**, and **46B2** which provide the fault action signals **86'**, **88'**, **90'**, and **92'** to all the gate terminals **48** of the inverters **30A** and **30B**.

(105) In some other embodiments and as shown in FIG. 5F, the three-terminal power semiconductor devices **42** of the bank **44** of both of the inverters **30A** and **30B** are caused to open and the three-terminal power semiconductor devices **42** of the bank **40** of both of the inverters **30A** and **30B** are caused to short. In such embodiments, the processors **64A** and **64B** generate the control signals **86** and **90** that are configured to turn off all the gate terminals **48** in the banks **44** of the inverters **30A** and **30B** and, as a result, cause all of the three-terminal power semiconductor devices **42** in the bank **44** of the inverters **30A** and **30B** to open. The processors **64A** and **64B** also generate the control signals **88** and **92** that are configured to turn on all the gate terminals **48** in the bank **40** of the inverters **30A** and **30B** and, as a result, cause all of the three-terminal power semiconductor devices **42** in the bank **40** of the inverters **30A** and **30B** to short. The control signals **86**, **88**, **90**, and **92** are provided to the buffers **94**, **96**, **98**, and **100**, respectively, and are, in turn, provided to the drives **46A1**, **46A2**, **46B1**, and **46B2** which provide the fault action signals **86'**, **88'**, **90'**, and **92'** to all the gate terminals **48** of the inverters **30A** and **30B**.

(106) In various embodiments and as also shown in FIGS. 5A and 5C, various faults associated with the inverters **30A** and **30B** are not monitored by the processors **64A** and **64B** for the inverters **30A** and **30B**, respectively—because such faults are faults in one or both of the processors **64A** and/or **64B** or circuitry (discussed below) associated with the processors **64A** and **64B**. For such faults, in various embodiments the same fault response is applied to the three-terminal power semiconductor devices **42** (FIG. 3A) of both of the inverters **30A** and **30B** by causing the three-terminal power semiconductor devices **42** (FIG. 3A) of one of the banks **40** or **44** of both of the inverters **30A** and **30B** to short. Because the functionality of the processors **64A** and **64B** is not verifiable in such fault conditions, it is not verifiable that the processors **64A** and **64B** are able to

receive and process information regarding speed of the motors (or back EMF or voltage of the battery **14**). As a result, a default fault action is entailed in such instances. In various embodiments, that default action is a three-phase short of the three-terminal power semiconductor devices **42** in one of the banks **44** or **40** of both of the inverters **30A** and **30B**.

(107) In various embodiments and as shown in FIG. 5A AND 5C, the controller **32** includes health monitoring circuitry **102** configured to monitor health of the processors **64A** and **64B** via data links **104** and **106**, respectively. In some such embodiments the health monitoring circuitry **102** includes a field programmable gate array (“FPGA”). In such embodiments, the health monitoring circuitry **102** may be programmed to perform a safety check, such as a safety handshaking check, and to monitor whether the processors **64A** and/or **64B** are functioning properly (or functioning at all). In some such embodiments, the health monitoring circuitry **102** may be programmed to implement a rolling counter to perform such monitoring of the processors **64A** and **64B**. In some such embodiments, the processors **64A** and **64B** also monitor whether the health monitoring circuitry **102** is functioning. As such, a fault in the health monitoring circuitry **102** is considered to be a fault in circuitry associated with the processors **64A** and **64B**.

(108) In various embodiments and as also shown in FIGS. 5A and 5C, the controller **32** includes three-phase short circuitry **108**. In such embodiments, the three-phase short circuitry **108** is configured to generate fault action signals (discussed below) for faults that are not monitored by the processors **64A** and **64B**— because the faults may include faults in the processors **64A** and/or **64B** and/or the health monitoring circuitry **102**. The fault action signals generated by the three-phase short circuitry **108** causes the same fault response to be applied to the three-terminal power semiconductor devices **42** (FIG. 3A) of both of the inverters **30A** and **30B** by causing the three-terminal power semiconductor devices **42** (FIG. 3A) of one of the banks **40** or **44** of both of the inverters **30A** and **30B** to short.

(109) In various embodiments, the three-phase short circuitry **108** is external to the processors **64A** and **64B** and is configured to apply a same fault action to the power inverter **30A** and the power inverter **30B** for applying equalized torque to each wheel **16** or **18** operatively coupled to the drive unit **20** responsive to a fault not monitored by the **64A** processor or the processor **64B**. As shown in FIGS. 5A and 5C, the three-phase short circuitry **108** is coupled to receive a control signal **110** from the health monitoring circuitry **102**, a control signal **112** from the processor **64A**, and a control signal **114** from the processor **64B**. The health monitoring circuitry **102** is configured to generate the control signal **110** responsive to receiving a processor fault indication signal, indicative of a fault within the processor **64A**, from the processor **64A** via the data link **104** or receiving a processor fault indication signal, indicative of a fault within the processor **64B**, from the processor **64B** via the data link **106**. The processors **64A** and **64B** are configured to generate the control signals **112** and **114**, respectively, responsive to detecting failure of the health monitoring circuitry **102** (such as, without limitation, failure of the rolling counter) via the data links **104** and **106**, respectively.

(110) In various embodiments and as shown in FIG. 5C, the three-phase short circuitry **108** includes a voltage regulator **116**, a buffer **118**, and a buffer **120**. In some such embodiments, the voltage regulator **116** includes a voltage regulator configured to convert 12 VDC to 5 VDC. In such embodiments, the control signals **110**, **112**, and **114** suitably are 12 VDC signals. In response to application of any of the 12 VDC control signals **110**, **112**, or **114**, the voltage regulator **116** outputs a 5 VDC control signal **122**. The control signal **122** is input to the buffers **118** and **120**. The buffer **118** is coupled to provide the control signal **122** to the drives **46A2** for the bank **44** of the power inverter **30A**. The drives **46A2** generate and provide the fault action signal **88'** to the gate terminals **48** of the bank **44** of the power inverter **30A**, thereby causing the three-terminal power semiconductor devices **42** of the bank **44** of the power inverter **30A** to short. The buffer **120** is coupled to provide the control signal **122** to the drives **46B2** for the bank **44** of the power inverter **30B**. The drives **46B2** generate and provide the fault action signal **92'** to the gate terminals **48** of

the bank **44** of the power inverter **30B**, thereby causing the three-terminal power semiconductor devices **42** of the bank **44** of the power inverter **30B** to short.

(111) It will be appreciated that shorting the three-terminal power semiconductor devices **42** of the banks **44** of the inverters **30A** and **30B** is given by way of illustration only and not of limitation. In some embodiments, the three-terminal power semiconductor devices **42** of the banks **40** of the inverters **30A** and **30B** are shorted (and the three-terminal power semiconductor devices **42** of the banks **44** of the inverters **30A** and **30B** remain open).

(112) Referring additionally to FIG. **5G**, in various embodiments an illustrative method **124** is provided for applying a same fault action to a first power inverter and a second power inverter for applying equalized torque to each wheel operatively coupled to a drive unit responsive to a fault associated with the first power inverter or the second power inverter.

(113) The method **124** begins at a block **126**. At a block **128** a fault associated with a first power inverter or a second power inverter of a drive unit of an electric vehicle is detected. At a block **130**, responsive to detecting the fault, a same fault action is applied to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit. The method **124** ends at a block **130**.

(114) In various embodiments, a first processor for the first power inverter and a second processor for the second power inverter may monitor for the fault.

(115) In various embodiments, the processor for the inverter with the fault associated therewith may communicate the fault action to the processor for the inverter without a fault associated therewith.

(116) In various embodiments, the fault action may include causing all banks of three-terminal power semiconductor devices in the first power inverter and the second power inverter to open for motor speed less than a threshold speed or causing one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short for motor speed greater than the threshold speed.

(117) In various embodiments, the fault associated with a first power inverter or a second power inverter of a drive unit of an electric vehicle may include a fault in a first processor for the first power inverter, or a second processor for the second power inverter, or health monitoring circuitry for the first processor and the second processor.

(118) In various embodiments, applying a same fault action to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit may include applying, by circuitry external to the first processor and the second processor, a same fault action to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit.

(119) In various embodiments, the fault action may include causing one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short.

(120) Notwithstanding the above, it will be appreciated that in some instances speed of one motor may be different from speed of another motor. For example, during a turn, an outside wheel might be rotating faster than an inside wheel. Similarly, wheels may be rotating at different speeds in various wheel slip situations. In some such instances, because the processor **64A** selects the appropriate speed-dependent fault action for its associated electrical motor **24A** and the processor **64B** selects the appropriate speed-dependent fault action for its associated electrical motor **24B**, a speed-dependent fault action for one motor of a drive unit **20** (for example, based on back EMF and voltage of the battery **14**) may be different from a speed-dependent fault action for the other motor of the drive unit **20**.

(121) In addition to the faults associated with the inverters **30A** and **30B** discussed above, loss of low voltage DC electrical power to the controller **32** (such as 12 VDC and referred to herein as 12V) results in a same fault action being applied to both inverters **30A** and **30B** in order to help avoid a torque imbalance between the two wheels **16** or **18** driven by a same drive unit **20**.

(122) It will be appreciated that, as shown in FIGS. 5A and 5C, the processors **64A** and **64B**, the drives **46A1**, **46A2**, **46B1**, and **46B2**, the health monitoring circuitry **102**, and the three-phase short circuitry **108** all include components that are powered by low voltage DC electrical power (12V) supplied to the controller **32**. It will be appreciated that loss of 12V supplied to the controller **32** means that the processors **64A** and **64B**, the drives **46A1**, **46A2**, **46B1**, and **46B2**, the health monitoring circuitry **102**, and the three-phase short circuitry **108** are not available to apply a same fault action to both inverters **30A** and **30B** as discussed above.

(123) Thus, in various embodiments and as discussed below, capability is provided to apply a same fault action to both inverters **30A** and **30B** in the event of loss of 12V supplied to the controller **32**.

(124) As shown in FIGS. 5A and 5C and referring additionally to FIGS. 6A-6B, in various embodiments and by way of overview a detection circuit **140** is configured to detect loss of low voltage DC electrical power (12V) supplied to the controller **32**. A backup power circuit **84A** is associated with the power inverter **30A** and a backup power circuit **84B** is associated with the power inverter **30B**. Each backup power circuit **84A** and **84B** is configured to convert high voltage DC electrical power, such as 450 VDC, to low voltage DC electrical power (such as, for example, a step-down DC-DC converter) responsive to detection of loss of low voltage DC electrical power supplied to the controller **32**. The three-phase short circuitry **108** is configured to apply a same fault action to the power inverter **30A** and the power inverter **30B** for applying equalized torque to each wheel **16** or **18** operatively coupled to the drive unit **20** responsive to detection of loss of low voltage DC electrical power supplied to the controller **32**.

(125) As shown in FIGS. 5A, 5C, and 6A, in various embodiments the detection circuit **140** uses an optocoupler **141** to detect loss of low voltage DC electrical power (12V) supplied to the controller **32**. Low voltage DC electrical power (such as 12V) is supplied to a resistor R.sub.1 which is electrically connected in series to a resistor R.sub.2. The resistors R.sub.1 and R.sub.2 function as a voltage divider. A control signal **143** at a suitable voltage (such as, without limitation 5V) is supplied from a node **145** between the resistors R.sub.1 and R.sub.2 to a light-emitting diode (LED) **145** of the optocoupler **141**. When energized, the LED **145** converts the electrical input into light and emits light (either visible light or infrared (IR) light). A phototransistor **147** detects the light emitted by the LED **145** and turns on. In various embodiments the phototransistor **149** is a pull-down transistor. The phototransistor **149** is coupled to provide an enable signal **151** to the backup power circuits **84A** and **84B**.

(126) During normal operation, the control signal **143** is supplied to the LED **147** and the LED **147** emits light. The phototransistor **149** detects the light and turns on. Because the phototransistor **149** is a pull-down transistor, the enable signal **151** is low when the phototransistor **149** is on. When the enable signal **151** is low, the enable signal **151** pulls down the backup power circuits **84A** and **84B** (such as step-down DC-DC converters) such that the backup power circuits **84A** and **84B** are turned off and are prevented from applying a three-phase short.

(127) In the event of loss of low voltage DC electrical power (12V), the signal **143** goes away and the LED **147** stops emitting light. As a result, the phototransistor **149** turns off and the enable signal **151** is high. When the enable signal **151** is high, the backup power circuits **84A** and **84B** are turned on and apply a three-phase short as described below. For example, the enable signal **151** may turn on a step-down DC-DC converter in each of the backup power circuits **84A** and **84B** to convert high voltage DC electrical power to low voltage DC electrical power (12V).

(128) In various embodiments the gate drive circuit **46A** and the gate drive circuit **46B** include the backup power circuits **84A** and **84B**, respectively. High voltage DC electrical power (such as 450V) is provided to each of the backup power circuits **84A** and **84B**. Each backup power circuit **84A** and **84B** is configured to convert 450V DC electrical power to low voltage DC electrical power, such as 12 VDC electrical power. As such, the backup power circuits **84A** and **84B** suitably include a step-down DC-DC converter (as discussed above), voltage divider circuitry (as discussed below) if desired, or the like.

(129) As shown in FIG. 6B, if desired, in some embodiments each backup power circuit **84A** and **84B** may optionally include a normally-open relay **152** having a relay coil **154** operably coupled to receive low voltage DC electrical power (12V) from the controller **32**. In such embodiments the normally-open relay **152** also includes a normally-open contact **156**. As discussed above, during normal operation low voltage DC electrical power 12V is provided to the relay coil **154** and the relay coil **154** is energized, thereby causing the normally-open contact **156** to open). In the event of loss of low voltage DC electrical power (12V), the relay coil **154** is de-energized, thereby causing the normally-open contact **156** to shut). Thus, in such embodiments, the detection circuit **140** suitably includes the relay coil **154** and the low voltage DC electrical power (12V) from the controller **32** may be considered to be a control signal.

(130) In such embodiments, each backup power circuit **84A** and **84B** suitably includes a voltage divider **158** operably coupled to receive the high voltage DC electrical power. The voltage divider **158** is configured to convert the high voltage DC electrical power to the low voltage DC electrical power and is further configured to output the low voltage DC electrical power to the normally-open contact **156**. In such embodiments, during normal operation (when low voltage DC electrical power is provided to the relay coil **154** and the normally-open contact **156** is open), the backup power circuits **84A** and **84B** do not provide 12V electrical power. Conversely, in the event of loss of low voltage DC electrical power (12V) (when the relay coil **154** is de-energized and the normally-open contact **156** is shut), the backup power circuits **84A** and **84B** provide 12V electrical power. Thus, in such embodiments, each backup power circuit **84A** and **84B** also suitably includes the normally-open contact **156**.

(131) As discussed above, in various embodiments the power inverter **30A** and the power inverter **30B** each include two banks **40** and **44** of the three-terminal power semiconductor devices **42**. As discussed below, the three-phase short circuitry **108** is further configured to cause one bank **40** or **44** of the three-terminal power semiconductor devices **42** in the power inverter **30A** and the power inverter **30B** to short responsive to detection of loss of low voltage DC electrical power supplied to the controller **32**.

(132) As shown in FIGS. 5A and 5C, regardless of how loss of low voltage DC electrical power (12V) is detected and how the low voltage DC electrical power (12V) is generated by the backup power circuits **84A** and **84B**, the low voltage DC electrical power (12V) is supplied from the backup power circuits **84A** and **84B** to the voltage regulator **116**. In response to application of the low voltage DC electrical power (12V), the voltage regulator **116** outputs a 5 VDC control signal **122**. The control signal **122** is input to the buffers **118** and **120**. The buffer **118** is coupled to provide the control signal **122** to the drives **46A2** for the bank **44** of the power inverter **30A**. The drives **46A2** generate and provide the fault action signal **88'** to the gate terminals **48** of the bank **44** of the power inverter **30A**, thereby causing the three-terminal power semiconductor devices **42** of the bank **44** of the power inverter **30A** to short. The buffer **120** is coupled to provide the control signal **122** to the drives **46B2** for the bank **44** of the power inverter **30B**. The drives **46B2** generate and provide the fault action signal **92'** to the gate terminals **48** of the bank **44** of the power inverter **30B**, thereby causing the three-terminal power semiconductor devices **42** of the bank **44** of the power inverter **30B** to short.

(133) It will be appreciated that shorting the three-terminal power semiconductor devices **42** of the banks **44** of the inverters **30A** and **30B** is given by way of illustration only and not of limitation. In some embodiments, the three-terminal power semiconductor devices **42** of the banks **40** of the inverters **30A** and **30B** are shorted (and the three-terminal power semiconductor devices **42** of the banks **44** of the inverters **30A** and **30B** remain open).

(134) In various embodiments and as shown in FIG. 6C, a method **160** is provided for applying a same fault action to both inverters of a drive unit of an electric vehicle in the event of loss of 12V supplied to a common controller for both inverters.

(135) The method **160** starts at a block **162**. At a block **164**, loss of low voltage direct current (DC)

electrical power supplied to a controller for a first processor for a first power inverter and a second processor for a second power inverter of a drive unit for an electric vehicle is detected. At a block **166** high voltage DC electrical power is converted to low voltage DC electrical power responsive to detection of loss of low voltage DC electrical power supplied to the controller. At a block **168** a same fault action is applied to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit responsive to detection of loss of low voltage DC electrical power supplied to the controller. The method **160** stops at a block **170**. (136) In various embodiments a control signal may be provided responsive to presence of low voltage DC electrical power supplied to the controller.

(137) In various embodiments applying a same fault action to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit responsive to detection of loss of low voltage DC electrical power supplied to the controller may include applying a same fault action to the first power inverter and the second power inverter for applying equalized torque to each wheel operatively coupled to the drive unit responsive to absence of the control signal.

(138) In various embodiments the fault action may include causing one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short.

(139) Those skilled in the art will recognize that at least a portion of the devices and/or processes described herein can be integrated into a data processing system. Those having skill in the art will recognize that a data processing system generally includes one or more of a system unit housing, a video display device, memory such as volatile or non-volatile memory, processors such as microprocessors or digital signal processors, computational entities such as operating systems, drivers, graphical user interfaces, and applications programs, one or more interaction devices (e.g., a touch pad, a touch screen, an antenna, etc.), and/or control systems including feedback loops and control motors (e.g., feedback for sensing position and/or velocity; control motors for moving and/or adjusting components and/or quantities). A data processing system may be implemented utilizing suitable commercially available components, such as those typically found in data computing/communication and/or network computing/communication systems.

(140) The term module, as used in the foregoing/following disclosure, may refer to a collection of one or more components that are arranged in a particular manner, or a collection of one or more general-purpose components that may be configured to operate in a particular manner at one or more particular points in time, and/or also configured to operate in one or more further manners at one or more further times. For example, the same hardware, or same portions of hardware, may be configured/reconfigured in sequential/parallel time(s) as a first type of module (e.g., at a first time), as a second type of module (e.g., at a second time, which may in some instances coincide with, overlap, or follow a first time), and/or as a third type of module (e.g., at a third time which may, in some instances, coincide with, overlap, or follow a first time and/or a second time), etc.

Reconfigurable and/or controllable components (e.g., general purpose processors, digital signal processors, field programmable gate arrays, etc.) are capable of being configured as a first module that has a first purpose, then a second module that has a second purpose and then, a third module that has a third purpose, and so on. The transition of a reconfigurable and/or controllable component may occur in as little as a few nanoseconds, or may occur over a period of minutes, hours, or days.

(141) In some such examples, at the time the component is configured to carry out the second purpose, the component may no longer be capable of carrying out that first purpose until it is reconfigured. A component may switch between configurations as different modules in as little as a few nanoseconds. A component may reconfigure on-the-fly, e.g., the reconfiguration of a component from a first module into a second module may occur just as the second module is needed. A component may reconfigure in stages, e.g., portions of a first module that are no longer needed may reconfigure into the second module even before the first module has finished its

operation. Such reconfigurations may occur automatically, or may occur through prompting by an external source, whether that source is another component, an instruction, a signal, a condition, an external stimulus, or similar.

(142) For example, a central processing unit of a personal computer may, at various times, operate as a module for displaying graphics on a screen, a module for writing data to a storage medium, a module for receiving user input, and a module for multiplying two large prime numbers, by configuring its logical gates in accordance with its instructions. Such reconfiguration may be invisible to the naked eye, and in some embodiments may include activation, deactivation, and/or re-routing of various portions of the component, e.g., switches, logic gates, inputs, and/or outputs. Thus, in the examples found in the foregoing/following disclosure, if an example includes or recites multiple modules, the example includes the possibility that the same hardware may implement more than one of the recited modules, either contemporaneously or at discrete times or timings. The implementation of multiple modules, whether using more components, fewer components, or the same number of components as the number of modules, is merely an implementation choice and does not generally affect the operation of the modules themselves. Accordingly, it should be understood that any recitation of multiple discrete modules in this disclosure includes implementations of those modules as any number of underlying components, including, but not limited to, a single component that reconfigures itself over time to carry out the functions of multiple modules, and/or multiple components that similarly reconfigure, and/or special purpose reconfigurable components.

(143) In some instances, one or more components may be referred to herein as “configured to,” “configured by,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that such terms (for example “configured to”) generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

(144) While particular aspects of the present subject matter described herein have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true spirit and scope of the subject matter described herein. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (for example, bodies of the appended claims) are generally intended as “open” terms (for example, the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (for example, “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (for example, the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in

general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

(145) The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, or examples can be implemented, individually and/or collectively, by a wide range of hardware, software (e.g., a high-level computer program serving as a hardware specification), firmware, or virtually any combination thereof, limited to patentable subject matter under 35 U.S.C. 101. In an embodiment, several portions of the subject matter described herein may be implemented via Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs), digital signal processors (DSPs), or other integrated formats. However, those skilled in the art will recognize that some aspects of the embodiments disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, limited to patentable subject matter under 35 U.S.C. 101, and that designing the circuitry and/or writing the code for the software (e.g., a high-level computer program serving as a hardware specification) and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as a program product in a variety of forms, and that an illustrative embodiment of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution. Examples of a signal bearing medium include, but are not limited to, the following: a recordable type medium such as a floppy disk, a hard disk drive, a Compact Disc (CD), a Digital Video Disk (DVD), a digital tape, a computer memory, etc.; and a transmission type medium such as a digital and/or an analog communication medium (e.g., a fiber optic cable, a waveguide, a wired communications link, a wireless communication link (e.g., transmitter, receiver, transmission logic, reception logic, etc.), etc.).

(146) With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flows are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

(147) While the disclosed subject matter has been described in terms of illustrative embodiments, it will be understood by those skilled in the art that various modifications can be made thereto without departing from the scope of the claimed subject matter as set forth in the claims.

Claims

1. A controller comprising: one or more processors associated with a first and second power inverter for a drive unit for an electric vehicle, the one or more processors comprising a first processor and a second processor; one or more computer-readable media for the one or more processors, each being configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter and based on common vehicle status signals and vehicle fault indications received at each of the one or more computer-readable media; and three-phase short circuitry, wherein: the same fault action further includes causing all of a plurality of banks of three-terminal power semiconductor devices in the first power inverter and the second power inverter to open when motor speed of the electric vehicle is less than a threshold speed, the fault includes a fault chosen from at least one of the first processor or the second processor and a fault in health monitoring circuitry configured to monitor health of at least one of the first processor or the second processor; and the three-phase short circuitry is configured to cause one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short responsive to the fault chosen from at least one of the first processor or the second processor and the fault in the health monitoring circuitry.
2. The controller of claim 1, wherein the computer-executable instructions are further configured to cause its associated processor to monitor for the fault.
3. The controller of claim 2, wherein the same fault action further includes and causing one bank of three-terminal power semiconductor devices in each of the first power inverter and the second power inverter to short when motor speed is greater than the threshold speed.
4. The controller of claim 1, further comprising a communications link, wherein the first processor is associated with the first power inverter for the drive unit and the second processor is associated with the second power inverter, and the communications link is between the first processor and the second processor.
5. The controller of claim 4, wherein the three-phase short circuitry is external to the first processor and the second processor and configured to apply a same fault action to the first power inverter and the second power inverter responsive to a fault not monitored by the first processor and the second processor, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.
6. The controller of claim 1, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.
7. A dual power inverter module comprising: a DC link capacitor electrically connectable to a source of high voltage DC electrical power; a first power inverter electrically connectable to the DC link capacitor and configured to convert the high voltage DC electrical power to three phase high voltage AC electrical power, the first power inverter being further configured to supply the three phase high voltage AC electrical power to a first electric motor; a second power inverter electrically connectable to the DC link capacitor and configured to convert the high voltage DC electrical power to three phase high voltage AC electrical power, the second power inverter being further configured to supply the three phase high voltage AC electrical power to a second electric motor; and a controller including: one or more processors associated with the first and second power inverters for a drive unit for an electric vehicle, the one or more processors comprising a first processor and a second processor; and one or more computer-readable media for the one or more processors, each being configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter

and the second power inverter and based on common vehicle status signals and vehicle fault indications received at the controller; and three-phase short circuitry, wherein: the same fault action further includes causing one bank of three-terminal power semiconductor devices in each of the first power inverter and the second power inverter to short when motor speed is greater than a threshold speed, the fault includes a fault chosen from at least one of the first processor or the second processor and a fault in health monitoring circuitry configured to monitor health of at least one of the first processor or the second processor; and the three-phase short circuitry is configured to cause one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short responsive to the fault chosen from at least one of the first processor or the second processor and the fault in the health monitoring circuitry.

8. The dual power inverter module of claim 7, wherein the computer-executable instructions are further configured to cause its associated processor to monitor for the fault.

9. The dual power inverter module of claim 8, wherein the fault action further includes an action chosen from causing all of a plurality of banks of three-terminal power semiconductor devices in the first power inverter and the second power inverter to open when motor speed is less than a threshold speed.

10. The dual power inverter module of claim 7, further comprising a communications link, wherein the first processor is associated with the first power inverter for the drive unit and the second processor is associated with the second power inverter, and the communications link is between the first processor and the second processor.

11. The dual power inverter module of claim 10, wherein the three-phase short circuitry is external to the first processor and the second processor and configured to apply a same fault action to the first power inverter and the second power inverter responsive to a fault not monitored by the first processor and the second processor, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

12. The dual power inverter module of claim 7, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

13. An electric vehicle comprising: a vehicle body; a high voltage direct current (DC) electrical battery and a low voltage DC electrical battery, wherein the high voltage DC electrical battery and the low voltage DC electrical battery are disposed within the vehicle body; first and second electric motors mechanically couplable to rotate at least one set of axles; and at least one dual power inverter module including: a DC link capacitor electrically connectable to the high voltage DC electrical battery to provide a source of high voltage DC electrical power; a first power inverter electrically connectable to the DC link capacitor and configured to convert the high voltage DC electrical power to three phase high voltage AC electrical power, the first power inverter being further configured to supply the three phase high voltage AC electrical power to the first electric motor; a second power inverter electrically connectable to the DC link capacitor and configured to convert the high voltage DC electrical power to three phase high voltage AC electrical power, the second power inverter being further configured to supply the three phase high voltage AC electrical power to the second electric motor; and a controller including: one or more processors associated with the first and second power inverters for a drive unit for the electric vehicle, the one or more processors comprising a first processor and a second processor; and one or more computer-readable media for the one or more processors, each being configured to store computer-executable instructions configured to cause its associated processor to apply a same fault action, based on a loss of power from the low voltage DC electrical battery to the controller, to the first power inverter and the second power inverter responsive to a fault associated with an inverter chosen from the first power inverter and the second power inverter and based on common vehicle status signals and vehicle fault indications received at the controller; and three-phase circuitry, wherein: the same fault action further includes causing all of a plurality of banks of three-terminal power semiconductor devices in the first power inverter and the second power inverter to open when

motor speed of the electric vehicle is less than a threshold speed the fault includes a fault chosen from at least one of the first processor or the second processor and a fault in health monitoring circuitry configured to monitor health of at least one of the first processor or the second processor; and the three-phase circuitry is configured to cause one bank of three-terminal power semiconductor devices in the first power inverter and the second power inverter to short responsive to the fault chosen from at least one of the first processor or the second processor and the fault in the health monitoring circuitry.

14. The electric vehicle of claim 13, wherein the computer-executable instructions are further configured to cause its associated processor to monitor for the fault.

15. The electric vehicle of claim 14, wherein the fault action further includes an action chosen from causing all of a plurality of banks of three-terminal power semiconductor devices in the first power inverter and the second power inverter to open when motor speed is less than a threshold speed and causing one bank of three-terminal power semiconductor devices in each of the first power inverter and the second power inverter to short when motor speed is greater than the threshold speed.

16. The electric vehicle of claim 13, further comprising a communications link, wherein the first processor is associated with the first power inverter for the drive unit and the second processor is associated with the second power inverter, and the communications link is between the first processor and the second processor.

17. The electric vehicle of claim 16, wherein the three-phase short circuitry is external to the first processor and the second processor and configured to apply a same fault action to the first power inverter and the second power inverter responsive to a fault not monitored by the first processor and the second processor, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.

18. The electric vehicle of claim 13, wherein the same fault action includes applying equalized torque to each axle operatively coupled to the drive unit.
